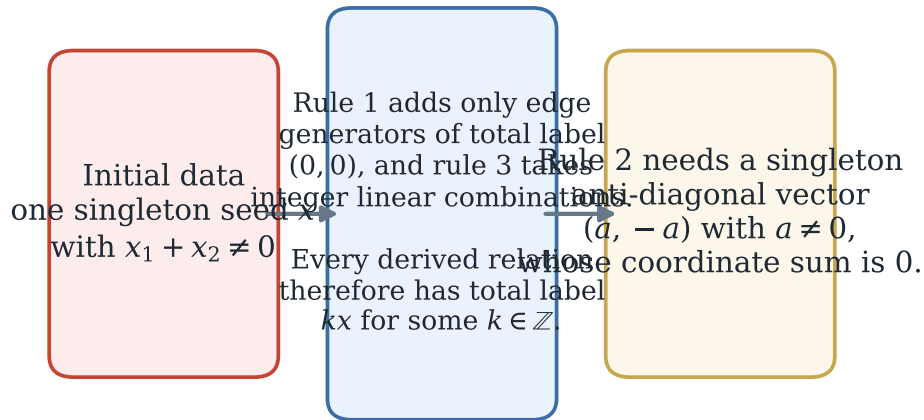


a

One-seed obstruction



If $kx = (a, -a)$ then $k(x_1 + x_2) = 0$. Because $x_1 + x_2 \neq 0$, we get $k = 0$, hence $(a, -a) = (0, 0)$, impossible.

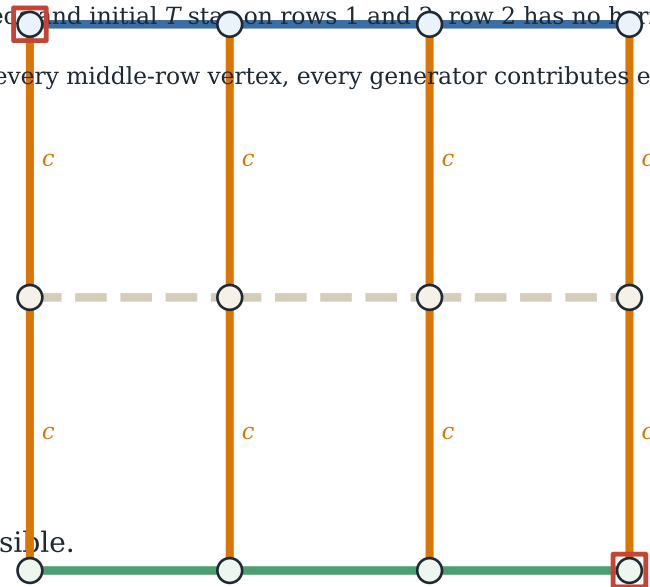
b

Row-quotient obstruction

Passive-middle $H = 3$ family

Seed and initial T start on rows 1 and 2; row 2 has no horizontal labels.

At every middle-row vertex, every generator contributes either 0 or $\pm c$.



Any middle-row singleton must lie in $\mathbb{Z}c$; if $c_1 + c_2 \neq 0$, it can never be anti-diagonal.